

```

55.         element[key] = re.sub(r'\[(I|V|X)+\]', '', value)
56.
57.     ELEMENTS[line[0]] = element # Map the atomic number to the element.
58.     ELEMENTS[line[1]] = element # Map the symbol to the element.
59.
60. print('Periodic Table of Elements')
61. print('By Al Sweigart al@inventwithpython.com')
62. print()
63.
64. while True: # Main program loop.
65.     # Show table and let the user select an element:
66.     print(''
67.           1  2  3  4  5  6  7  8  9  10 11 12 13 14 15 16 17 18
68.           1 H
69.           2 Li Be
70.           3 Na Mg
71.           4 K  Ca Sc Ti V  Cr Mn Fe Co Ni Cu Zn Ga Ge As Se Br Kr
72.           5 Rb Sr Y  Zr Nb Mo Tc Ru Rh Pd Ag Cd In Sn Sb Te I  Xe
73.           6 Cs Ba La Hf Ta W  Re Os Ir Pt Au Hg Tl Pb Bi Po At Rn
74.           7 Fr Ra Ac Rf Db Sg Bh Hs Mt Ds Rg Cn Nh Fl Mc Lv Ts Og
75.
76.           Ce Pr Nd Pm Sm Eu Gd Tb Dy Ho Er Tm Yb Lu
77.           Th Pa U  Np Pu Am Cm Bk Cf Es Fm Md No Lr'')
78.     print('Enter a symbol or atomic number to examine, or QUIT to quit.')
79.     response = input('> ').title()
80.
81.     if response == 'Quit':
82.         sys.exit()
83.
84.     # Display the selected element's data:
85.     if response in ELEMENTS:
86.         for key in ALL_COLUMNS:
87.             keyJustified = key.rjust(LONGEST_COLUMN)
88.             print(keyJustified + ': ' + ELEMENTS[response][key])
89.             input('Press Enter to continue...')

```

Exploring the Program

Try to find the answers to the following questions. Experiment with some modifications to the code and rerun the program to see what effect the changes have.

1. What bug do you cause if you change `response == 'Quit'` on line 81 to `response == 'quit'`?
2. What happens if you delete or comment out lines 53 and 55?